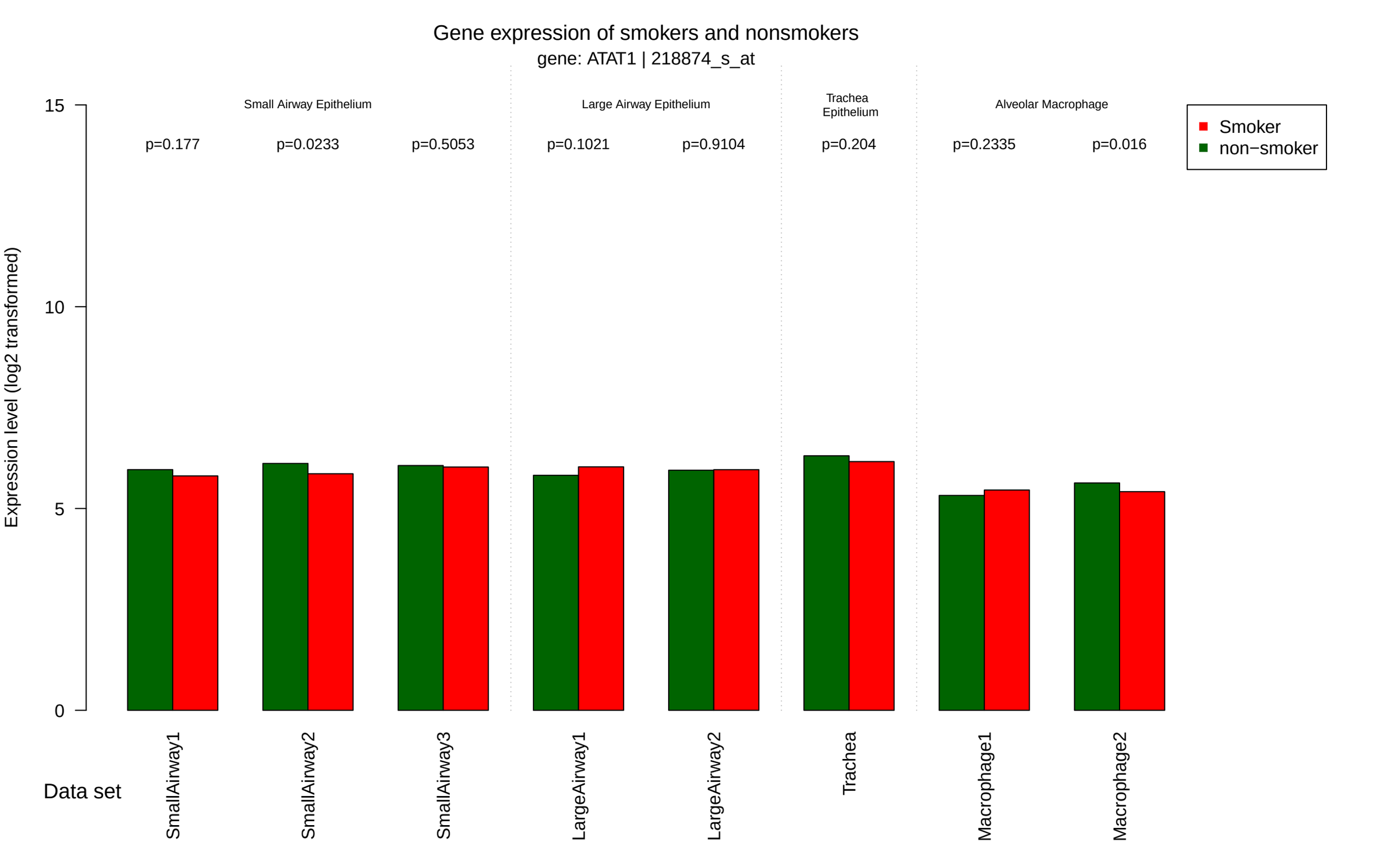
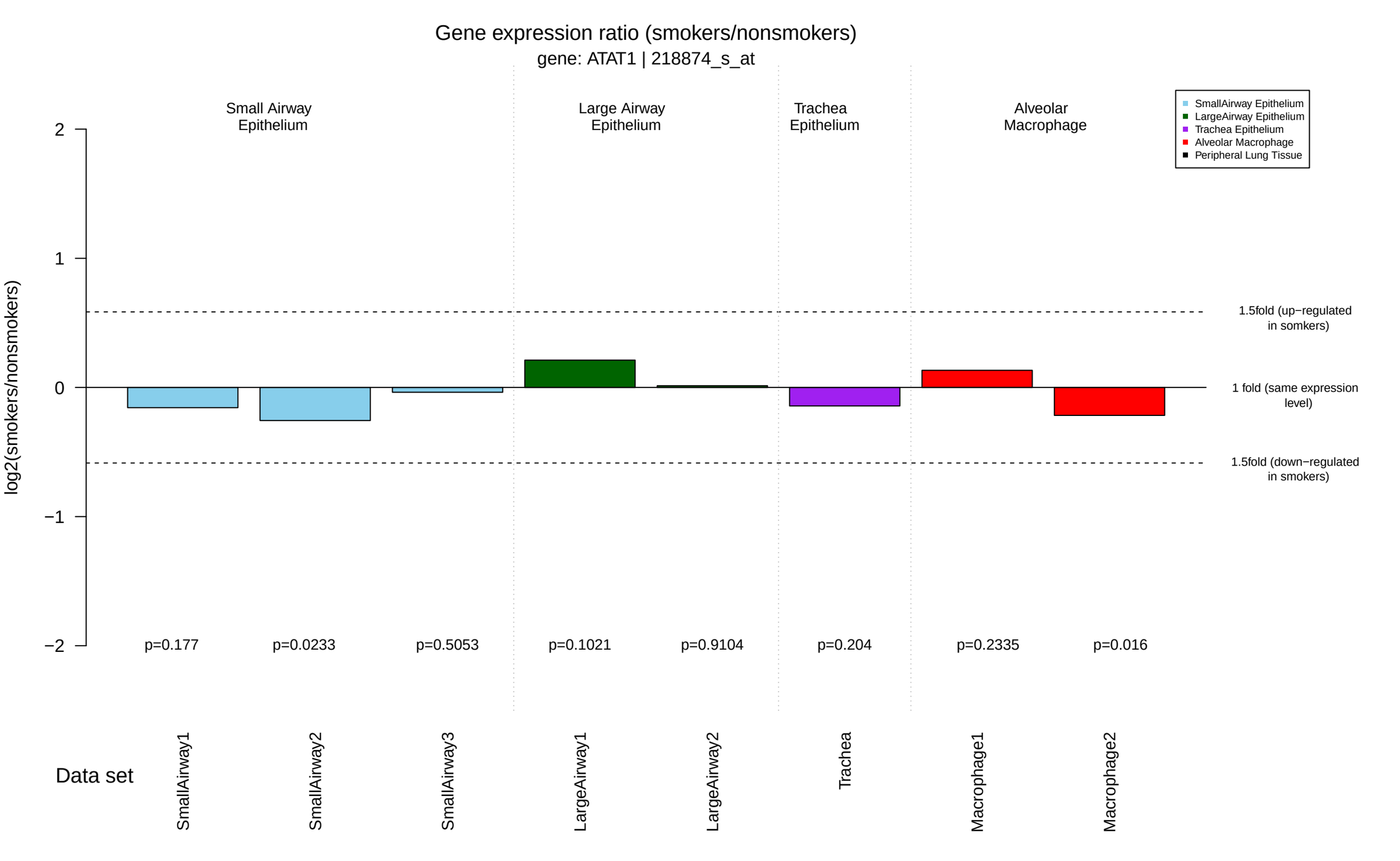
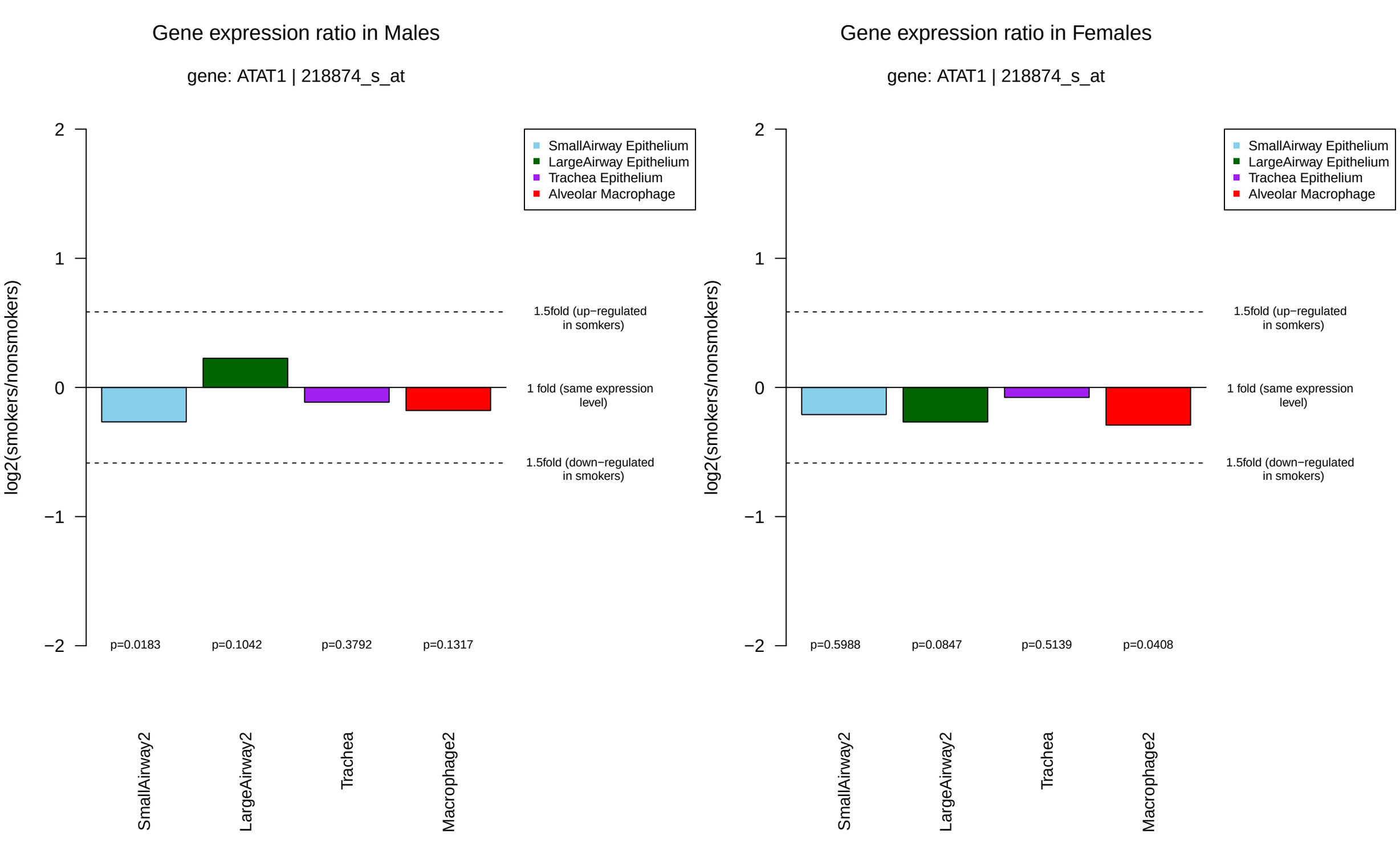
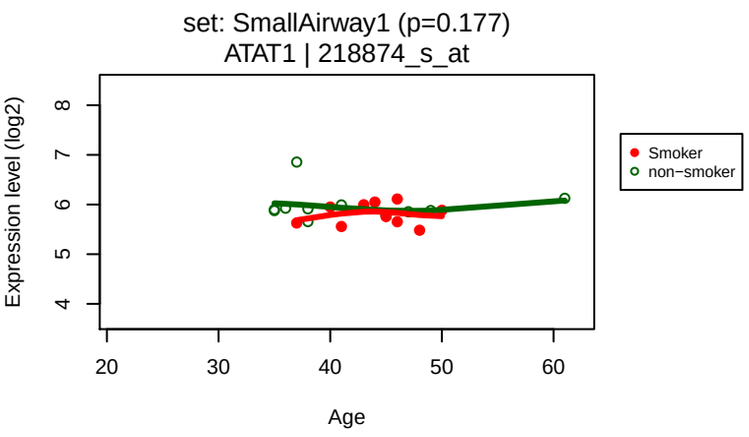
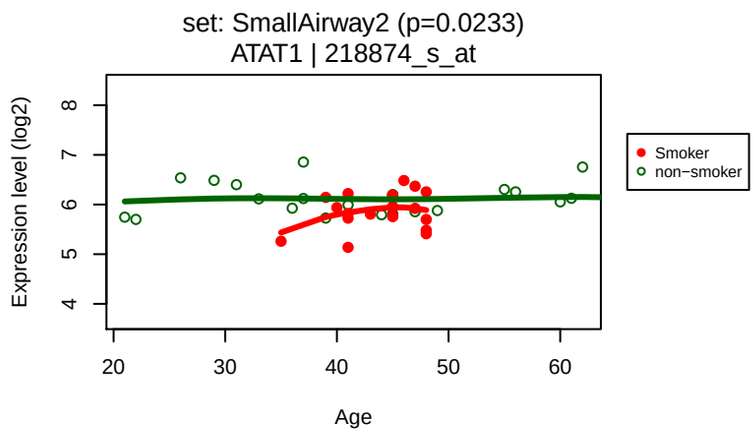
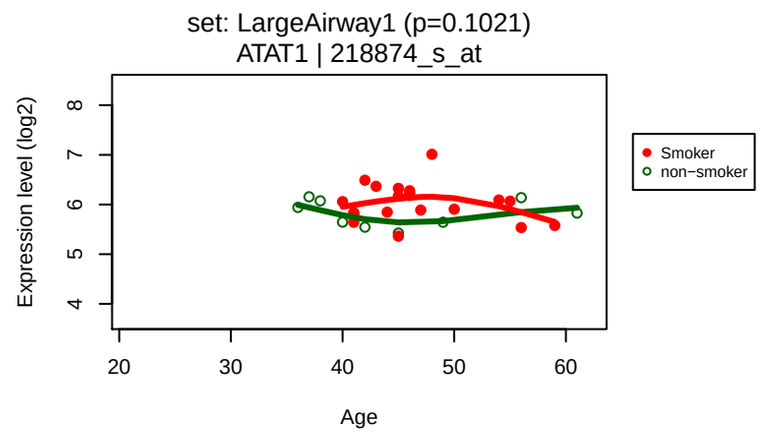

Gene expression of smokers and nonsmokers
gene: ATAT1 | 218874_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.177
p=0.0233
p=0.5053
p=0.1021
p=0.9104
p=0.204
p=0.2335
p=0.016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

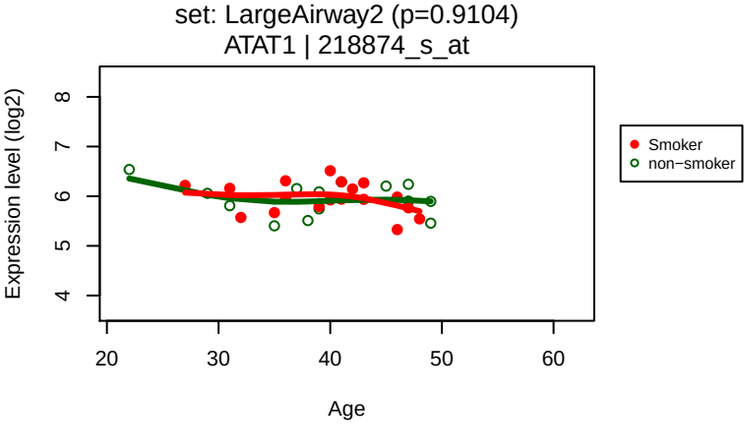

Gene expression ratio (smokers/nonsmokers)
gene: ATAT1 | 218874_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.177
p=0.0233
p=0.5053
p=0.1021
p=0.9104
p=0.204
p=0.2335
p=0.016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

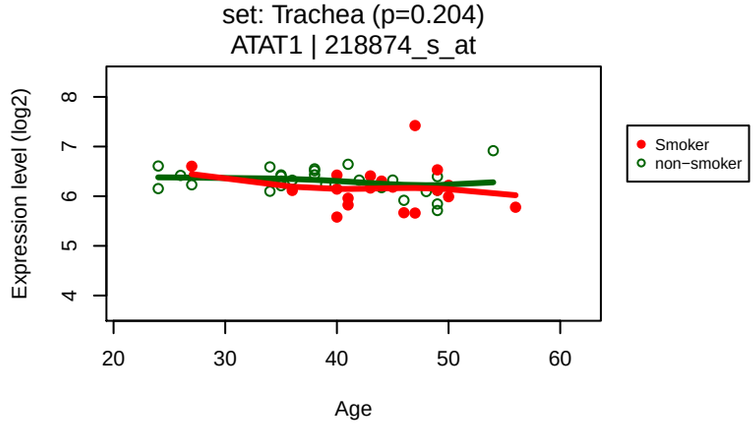

Gene expression ratio in Males
gene: ATAT1 | 218874_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0183
p=0.1042
p=0.3792
p=0.1317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATAT1 | 218874_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5988
p=0.0847
p=0.5139
p=0.0408
SmallAirway2
LargeAirway2
Trachea
Macrophage2

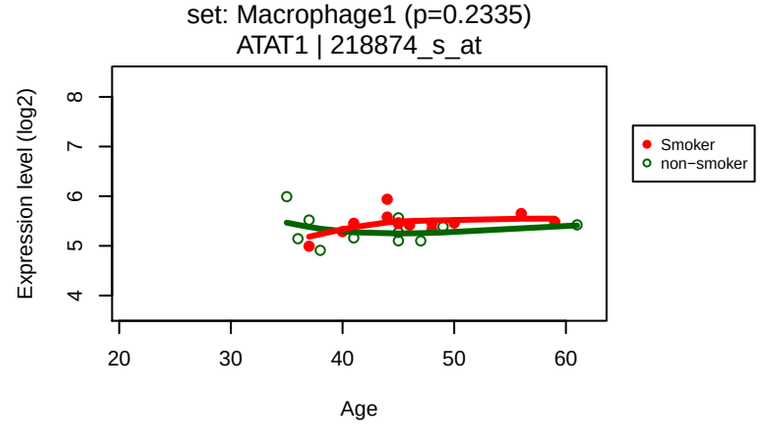

set: SmallAirway1 (p=0.177)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker

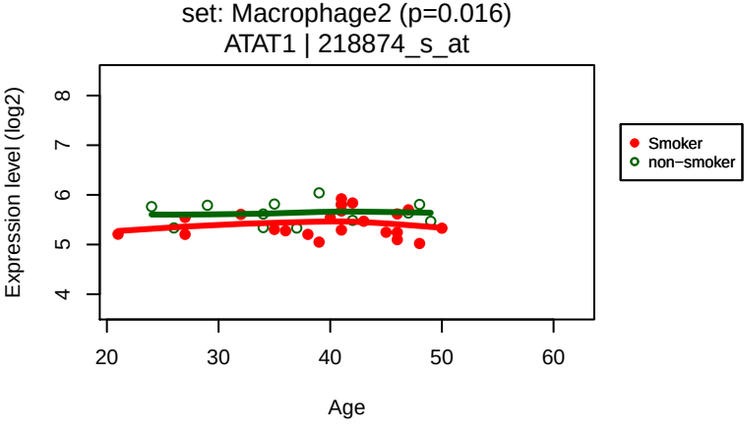

set: SmallAirway2 (p=0.0233)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker

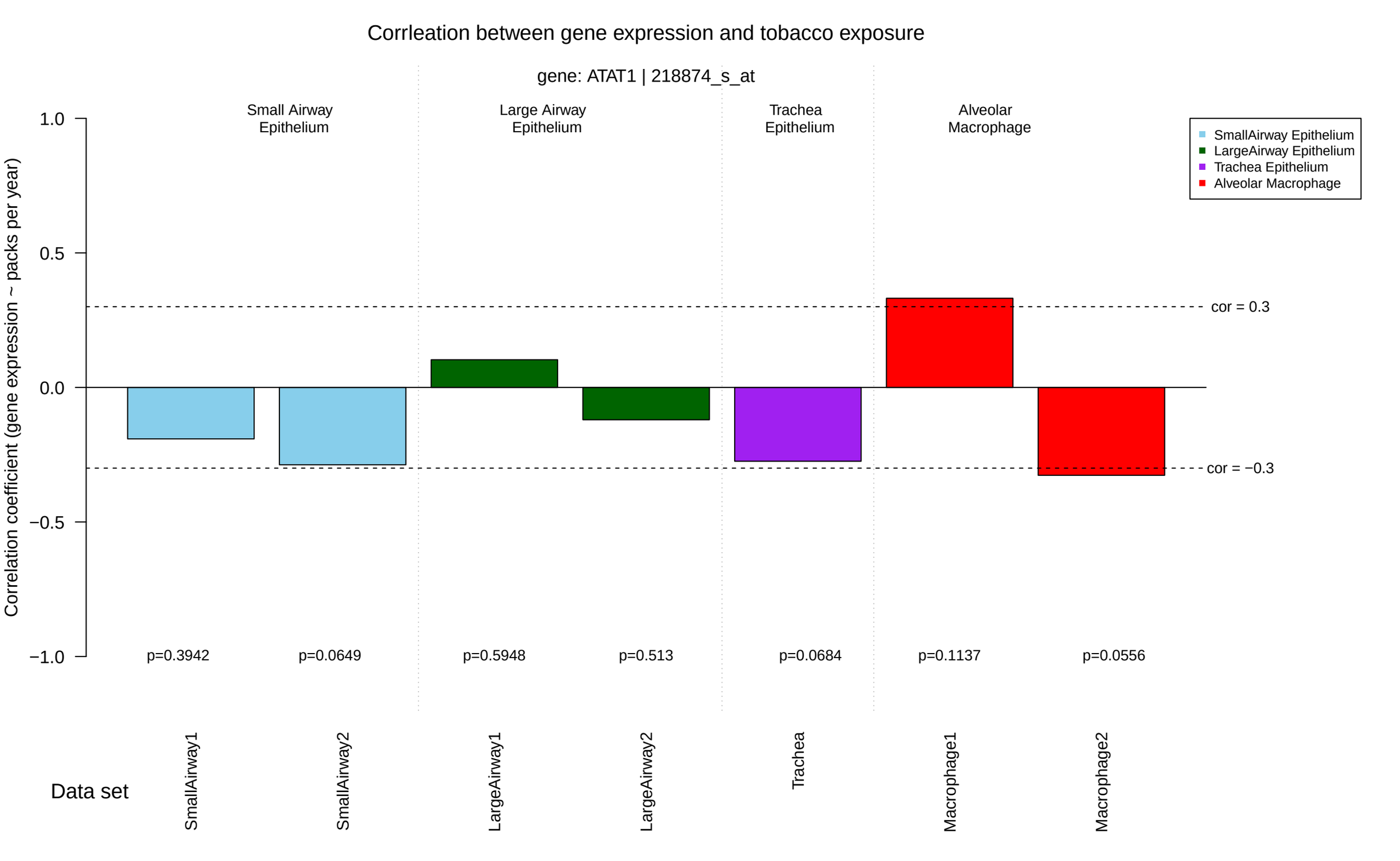

set: LargeAirway1 (p=0.1021)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9104)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.204)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2335)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.016)
ATAT1 | 218874_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATAT1 | 218874_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3942
p=0.0649
p=0.5948
p=0.513
p=0.0684
p=0.1137
p=0.0556
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2